

Evaluating Cross-Dataset Generalization of Efficient 3D CNN Architectures for Cardiac Substructure Segmentation

Digonta Das, Mihir Das, Sultan Helal Uddin Ahmed
Dept. of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh

Abstract—Automated cardiac segmentation is essential for diagnosing cardiovascular diseases, yet standard 3D Convolutional Neural Networks (CNNs) remain computationally expensive and prone to domain shift when deployed across different MRI scanners. This paper proposes an “Efficient SE-Mobile-Unet,” a lightweight hybrid architecture designed for resource-constrained clinical settings. By integrating MobileNetV2 blocks with Squeeze-and-Excitation (SE) attention mechanisms, we reduce the model size to approximately 2.2 million parameters while maintaining high accuracy. We validate our approach using a rigorous cross-vendor protocol: training on the ACDC dataset (Siemens/Philips) and performing zero-shot evaluation on the unseen M&Ms dataset (GE/Canon). Our model achieves a Dice Similarity Coefficient (DSC) of 0.8712 on the source domain and 0.7512 on the target domain, demonstrating a generalization gap of only 0.12 without retraining.

Index Terms—Cardiac MRI Segmentation, Efficient 3D CNN, Domain Generalization, MobileNetV2, Squeeze-and-Excitation.

I. INTRODUCTION

Cardiovascular diseases (CVDs) are a leading cause of mortality globally. While Cardiac MRI provides the high-resolution data needed to assess heart substructures—specifically the Left Ventricle (LV), Right Ventricle (RV), and Myocardium (Myo)—manual segmentation is a significant clinical bottleneck, often taking 20–30 minutes per patient.

Automated solutions using standard 3D CNNs (e.g., V-Net, 3D U-Net) often exceed 15 million parameters, requiring high-end GPUs with large VRAM [11]. Furthermore, these models suffer from “domain shift,” failing to generalize across MRI scanners from different vendors (e.g., Siemens vs. GE) due to variations in magnetic field strength and resolution [1]. This work presents a lightweight, hardware-aware architecture capable of robust cross-vendor generalization suitable for deployment on standard hardware.

II. METHODOLOGY

Our proposed framework is a fully automated end-to-end pipeline consisting of four stages: Input, Preprocessing, the Efficient 3D CNN Core, and Segmentation Output.

A. Data Preprocessing: The First Line of Defense

To mitigate domain shift caused by heterogeneous MRI data, we implemented a unified preprocessing pipeline crucial for minimizing variance.

1) *Intensity Normalization*: MRI signal intensities do not have a standard scale (unlike CT Hounsfield units). We apply Z-score normalization to each volume V to harmonize contrast and brightness:

$$V_{norm} = \frac{V - \mu}{\sigma} \quad (1)$$

where μ is the mean intensity and σ is the standard deviation of the volume.

2) *Spatial Resampling*: To ensure the network processes anatomically consistent shapes and fits within the 12GB VRAM limit of our test hardware, all volumes are resampled to fixed spatial dimensions of $16 \times 128 \times 128$ voxels.

B. Architecture: Efficient SE-Mobile-Unet

We propose a hybrid architecture that retains the classic encoder-decoder topology of the 3D U-Net to preserve high-resolution spatial details while replacing the heavy encoder blocks with lightweight alternatives [4].

1) *Depthwise Separable Convolutions*: To address the “Efficiency” goal, we replace standard 3D convolutions with Depthwise Separable Convolutions (DSC). This decomposes the operation into two simpler steps:

- 1) **Depthwise Convolution**: Applies a single spatial filter to each input channel independently.
- 2) **Pointwise Convolution**: Applies a $1 \times 1 \times 1$ convolution to combine the outputs of the depthwise layer.

This decomposition reduces the computational cost by a factor of 8–9x compared to standard convolutions, resulting in a total parameter count of only ≈ 2.2 million.

2) *Squeeze-and-Excitation (SE) Attention*: To address the “Robustness” goal, we integrate SE Attention blocks. These blocks explicitly model the inter-dependencies between channels. The SE block acts as a “relevance filter,” learning to suppress less informative features (such as scanner-specific noise or artifacts) and emphasize relevant anatomical features.

3) *Network Topology*: The network structure follows a U-shaped design:

- **Encoder**: Consists of three stages of MobileNet blocks combined with SE Attention. Filter counts increase progressively: 32, 64, and 128 filters.
- **Bottleneck**: A central layer utilizing MobileNet + SE with 256 filters.
- **Decoder**: Uses up-sampling followed by concatenation with skip connections from the encoder to recover spatial resolution.

III. EXPERIMENTAL SETUP

A. Multi-Source Evaluation Strategy

We employed a rigorous "Stress Test" strategy to evaluate generalization:

- **Source Domain (Training)**: ACDC Dataset (University Hospital of Dijon). Scanners: Siemens 1.5T and Philips 3.0T.
- **Target Domain (Zero-Shot Testing)**: M&Ms Dataset [1]. Scanners: General Electric (GE) and Canon. The model was evaluated on this dataset without any retraining.
- **Cross-Modality Test**: MM-WHS Dataset (CT and MRI) to test transferability across imaging modalities [3].

B. Implementation Details

The model was trained on an NVIDIA GeForce RTX 3060 (12GB VRAM).

- **Optimizer**: Adam (Learning rate = 0.001).
- **Loss Function**: A composite loss of Dice Loss + Cross-Entropy Loss was used to handle severe class imbalance and ensure sharp boundaries.
- **Epochs**: 10.

IV. RESULTS AND DISCUSSION

A. Quantitative Analysis

The model achieved robust segmentation performance across all substructures on the validation set.

TABLE I
OVERALL SEGMENTATION PERFORMANCE

Structure	Dice Score (F1)	IoU (Jaccard)
Right Ventricle (RV)	0.8214	0.7002
Myocardium (Myo)	0.7945	0.6610
Left Ventricle (LV)	0.9226	0.8570
Average	0.8462	0.7394

B. Cross-Vendor Generalization Gap

A key finding of this study is the minimized generalization gap. As shown in Table II, the performance drop when moving from the source domain (ACDC) to the unseen target domain (M&Ms) was limited.

TABLE II
CROSS-DATASET GENERALIZATION RESULTS

Dataset	Vendor	Dice Score	Status
ACDC	Siemens/Philips	0.8712	Source (Train)
M&Ms	GE/Canon	0.7512	Target (Zero-Shot)
Generalization Gap:		0.1200	

C. Qualitative Analysis

Visual inspection of error maps (comparing Prediction vs. Ground Truth) indicates that most errors are concentrated at the fine boundaries between structures, particularly the myocardium. This suggests that while the architecture is robust to domain shift, future work could benefit from boundary-aware loss functions.

V. CONCLUSION

We demonstrated that an efficient 3D CNN ($\approx 2.2M$ parameters) can achieve competitive accuracy and generalize across unseen scanner vendors with a Zero-Shot Dice score of 0.7512. The combination of MobileNetV2 blocks and SE Attention proves to be a viable strategy for deploying automated cardiac diagnostics in resource-constrained clinical environments.

REFERENCES

- [1] V. M. Campello *et al.*, "Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation: The M&Ms Challenge," *IEEE Trans. Med. Imaging*, vol. 40, no. 12, pp. 3543–3554, Dec. 2021, doi: 10.1109/TMI.2021.3090082.
- [2] Y. Zhao *et al.*, "Reverse Imaging for Wide-spectrum Generalization of Cardiac MRI Segmentation," in *Proc. Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
- [3] F. Wu and X. Zhuang, "Unsupervised Domain Adaptation with Variational Approximation for Cardiac Segmentation," *IEEE Trans. Med. Imaging*, 2021.
- [4] F. P. *et al.*, "ES-UNet: Efficient 3D Medical Image Segmentation with Enhanced Skip Connections," *NIH/PubMed*, 2024.
- [5] H. Ghouse, M. Alsharqi, F. R. Nezami, and M. Behzad, "PULSE: A unified multi-task architecture for cardiac segmentation, diagnosis, and few-shot cross-modality clinical adaptation," *arXiv preprint arXiv:2512.03848*, 2025.
- [6] Y. Zhou, J. Wu, W. Liao, S. Zhang, S. Zhang, and G. Wang, "TEGDA: Test-time evaluation-guided dynamic adaptation for medical image segmentation," in *Proc. Medical Image Computing and Computer Assisted Intervention (MICCAI)*, vol. 15965, *Lecture Notes in Computer Science (LNCS)*. Cham, Switzerland: Springer Nature, Sep. 2025, pp. 628–637.
- [7] F. Thaler, D. Štern, G. Plank, and M. Urschler, "Augmentation-based domain generalization and joint training from multiple source domains for whole heart segmentation," in *Comprehensive Analysis and Computing of Real-World Medical Images*, Cham, Switzerland: Springer Nature, 2025, pp. 168–179, doi: 10.1007/978-3-031-87009-5_17.
- [8] R. Safdari, M.-A. Nikouei Mahani, M. Koohi-Moghadam, and K. T. Bae, "MixStyleFlow: Domain generalization in medical image segmentation using normalizing flows," in *Proc. Medical Image Computing and Computer Assisted Intervention (MICCAI)*, vol. 15962, *Lecture Notes in Computer Science (LNCS)*. Cham, Switzerland: Springer Nature, Sep. 2025, pp. 376–385.
- [9] M. R. Casu and L. Carloni, "A machine learning approach to optimizing CNN deployment on tile-based systems-on-chip," M.S. thesis, Polytechnic Univ. of Turin, Turin, Italy, 2024. [Online]. Available: Webthesis.
- [10] N. Ravi *et al.*, "Accelerating 3D deep learning with PyTorch3D," *arXiv preprint arXiv:2007.08501*, 2020.
- [11] T. O. Ilesanmi, A. E. Ilesanmi, and B. O. Ajayi, "Reviewing 3D convolutional neural network approaches for medical image segmentation," *ScienceDirect*, 2024.

- [12] S. Tiwari, G. Jain, D. K. Shetty, M. Sudhi, J. M. Balakrishnan, and S. R. Bhatta, "A Comprehensive Review on the Application of 3D Convolutional Neural Networks in Medical Imaging," *Eng. Proc.*, vol. 59, no. 1, Art. no. 3, Dec. 2023, doi: 10.3390/engproc2023059003.